

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and applied quantitative scientist with experience developing and validating predictive and statistical models for financial risk, insurance pricing, and large, complex datasets. Builds reproducible Python/PyTorch and LightGBM workflows, designs rigorous performance and reliability evaluations, and translates research findings into implemented analyses and clear recommendations for business stakeholders.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Machine Learning: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), Linux, Git, AWS EC2, HPC/SLURM; supervised learning, GLM/GBM, feature engineering, feature attribution, numerical optimization, reproducible model-development workflows

Statistical Modeling and Validation: Probability, regression (GLM/GLS), hypothesis and sequential testing, Monte Carlo simulation, model selection and validation, predictive-performance evaluation, sensitivity analysis, missing-data and model-misspecification diagnostics

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Developed and validated a Sequential Probability Ratio Test framework for binary risk indicators, modeling the cumulative log-likelihood-ratio process as a random walk with statistically controlled decision boundaries.
- Applied Brownian-motion approximation and dynamic-programming probability propagation to evaluate boundary-crossing probabilities, stopping-time distributions, Type I/II error, and early-decision tradeoffs, documenting model behavior and limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ property-risk records, including data preparation, feature processing, GLM benchmarking, model validation, and predictive-performance evaluation on AWS EC2.
- Analyzed model performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

APPLIED STATISTICAL RESEARCH

Ph.D. Dissertation Research - Scalable Model Development and Reliability Diagnostics

Sep 2024 - Present

- **Model Development:** Developed a scalable Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- **Validation and Reliability:** Compared Matérn and generalized Cauchy specifications and designed scale- and frequency-resolved tests to identify misspecification in smoothness, noise, and dependence.
- **Data Quality and Robustness:** Built workflows for observation masking, quality filtering, missing-data handling, nearest-neighbor construction, dependency-aware ordering, and irregular-grid resampling; quantified the effect of data imperfections on estimates.
- **Research Engineering:** Implemented reusable Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, Monte Carlo simulation, CPU/GPU execution, cached artifacts, and restartable HPC jobs.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Linear Algebra, Statistical Computing, Stochastic Processes, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)